

DANIEL VAN LEWEN

Boston, MA | 209-380-9779 | danielvl@bu.edu | Portfolio: dvanlewen.github.io/ | LinkedIn: linkedin.com/in/daniel-van-lewen/

SUMMARY

PhD graduate in Mechanical Engineering specializing in robotics, hardware-software integration, and medical robot development. Experienced in real-time, autonomous navigation, actuator/sensor design, software architecture development, and experimental validation of soft robotic platforms. Passionate about solving human-centered problems through advanced robotic technologies.

TECHNICAL SKILLS

- **Programming Languages:** Python, MATLAB, C++, Arduino
- **Robotics:** ROS2, A*/Path Planning, OpenCV/Computer Vision, Machine Learning
- **Software:** CAD (Solidworks), 3DSlicer, Git, Labview, Microsoft Office Suite, Abaqus Finite Element Analysis, Minitab
- **Hardware:** 3D printing, experimental design, polymer injection molding, soft lithography, soldering, machining

EDUCATION

PhD, Mechanical Engineering, Boston University **Boston, MA**
Thesis: Soft Robots for Interventional Endoscopy 2020 - June 2025

MS, Mechanical Engineering, Boston University **Boston, MA**
Concentration: Dynamics, Systems, and Controls 2020-2022
Relevant Coursework: Robot Motion Planning, Machine Learning, Soft Robotics, Medical Robotics, Applied Math in Mechanics

BS, Mechanical Engineering, University of California, Los Angeles **Los Angeles, CA**
GPA: 3.6 2016 - 2020

PROFESSIONAL EXPERIENCE

Graduate Student Researcher | Material Robotics Lab **Sep 2020 - Present | Boston, MA**

- Led a cross-functional team of clinicians and engineering students on two soft surgical robotics projects, resulting in 6 peer-reviewed publications.
- Designed ROS2 software architecture for high accuracy (1 mm), semi-autonomous navigation of a lung needle biopsy robot.
- Integrated real-time feedback control with 3D spatial tracking and machine learning-enabled image detection.
- Built electro-mechanical and electro-pneumatic equipment for control of custom soft robots.
- Created test scripts to simulate and troubleshoot behavior of robot navigation systems.
- Designed navigation and signal processing algorithms for autonomous control and data monitoring in low latency (<100 ms), real-time systems.
- Integrated algorithms to interface commercial (UR5) robotic arms and custom actuation systems with vision and EM pose sensors.
- Developed benchtop, in-vitro, and ex-vivo (animal model) experiments for system validation.
- Supported patent writing for neurosurgical retraction projects.

Undergraduate Research Assistant | Bionics Lab **Sep 2018 - May 2020 | Los Angeles, CA**

- Designed and prototyped components for hand exoskeleton used in stroke patient rehabilitation.
- Sourced and assembled electronic and mechanical components on PCBs for control unit.
- Created 3D models and engineering drawings for manufacturers.

Engineer Intern | Parker Hannifin Corp. **Jul 2018 - Sep 2018; Jun 2017 - Sep 2017 | Modesto, CA**

- Designed experimental setups for fuel filter testing (burst pressure and life cycle analysis)
- Revised and created drawings for customer specifications
- Designed, tested, and implemented prototype fittings to address fuel filter air intrusion

PUBLICATIONS

1. **Van Lewen D.**, et al., "A Real-Time, Semi-Autonomous Navigation Platform for Soft Robotic Bronchoscopy", *IEEE Robotics and Automation Letters*, 2025.
2. **Van Lewen D.**, et al., "Capacitive Origami Sensing Modules for Measuring Force in a Neurosurgical, Soft Robotic Retractor", *IEEE International Conference on Robotic and Automation (ICRA)*, 2024
3. **Van Lewen D.**, et al., "A Millimeter-Scale Soft Robot for Tissue Biopsy Procedures", *Advanced Intelligent Systems*, 2023.
4. Amadeo T., **Van Lewen D.**, et al., "Soft Robotic Deployable Origami Actuators for Neurosurgical Brain Retraction", *Frontiers in Robotics and AI*, 2022.

CONFERENCE PRESENTATIONS

1. "Monitoring Forces in Soft Robotic Brain Retraction via Origami Sensing Modules", *Hamlyn Symposium on Medical Robotics*, 2024.
2. "A Fluidic Actuated Soft Robot for Improving Bronchoscopic Biopsy", *Hamlyn Symposium on Medical Robotics*, 2023.
3. "Soft Robotic Capacitive Sensing for Force Measurement in Neurosurgical Procedures", *Materials Research Society - Fall Meeting*, 2023.
4. "Actively Increasing Force Transmission of a Millimeter-Scale Soft Robot for Tissue Interactions in Minimally Invasive Surgery", *Materials Research Society - Spring Meeting*, 2023.

MENTORING AND TEACHING

- Graduate Student Teacher, Boston University: Product Design (2021), Heat Transfer (2022), Medical Robotics (2023, 2025).
- Mentored 10 undergraduate students, 1 high school student, and 4 graduate students in research projects.

PROFESSIONAL MEMBERSHIPS AND SERVICE

- Department of Education GAANN Fellow
- Member Institute of Electrical and Electronic Engineers (IEEE)
- Member IEEE Robotics and Automation Society
- Member of Material Research Society
- Reviewer for IEEE R-AL, ICRA, and Robosoft